“Everything is a subtle issue for you people! No wonder you never make any progress.”

“Now, now, Dad, we are getting at something significant here. What Popper was ultimately trying to pinpoint is that a good scientific theory has two characteristics. First, it is *definite*: you can’t just twist the theory to ‘explain’ anything at all, as Popper feared you could do with dialectical materialism or psychoanalysis. Second, it is *empirical*: theories are not deemed true by sheer reason alone. Rather, one imagines many different possible ways the world could be, each corresponding to a different theory, and then one chooses among the theories by going out and actually looking at the world.”

“Exactly.” Her father seemed to think that the advantage was his on this one. “Empirical! But if you can’t actually observe those worlds, there’s nothing really empirical about your theory at all.”

“Au contraire,” Alice replied. “Many-Worlds embodies both of these features perfectly. It is not a just-so story that can be adapted to any observed set of facts. Its postulates are simple: the world is described by a quantum wave function that evolves according to the Schrödinger equation. Those postulates are eminently falsifiable. Just do an experiment showing that quantum interference doesn’t occur when it should, or that entanglement really can be used for superluminal communication, or that a wave function really does collapse even without decoherence. Many-Worlds is the most falsifiable theory ever invented.”

“But those aren’t tests of Many-Worlds,” her father protested, unwilling to concede ground on this one. “Those are just tests of quantum mechanics generally.”

“Right! But Everettian quantum mechanics is just pure, austere quantum mechanics without any additional ad hoc assumptions. If you do want to introduce extra assumptions, then by all means we can ask whether those new assumptions are testable.”

“Come now. The defining feature of Many-Worlds is the existence of all those worlds out there. Our world can’t interact with them, so that particular aspect of the theory is untestable.”

“So what? Every good theory makes some predictions that are untestable. Our current theoretical understanding of general relativity predicts that the force of gravity will not tomorrow suddenly turn off for a period of one millisecond in a particular region of space ten meters across